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Ringraziamenti

...thanks to...

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Abstract

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Sommario è un breve riassunto del lavoro svolto dove si descrive l'obiettivo, l'oggetto della tesi, le metodologie e le tecniche usate, i dati elaborati e la spiegazione delle conclusioni alle quali siete arrivati.

Il sommario dell'elaborato consiste al massimo di 3 pagine e deve contenere le seguenti informazioni:

- contesto e motivazioni
- breve riassunto del problema affrontato
- tecniche utilizzate e/o sviluppate
- risultati raggiunti, sottolineando il contributo personale del laureando/a

1 Introduction and Background

1.1 Introduction

The global agricultural sector is currently facing unprecedented challenges. Driven by rapid population growth, climate change, and the need for sustainable resource management, modern agriculture must maximize crop yields while minimizing environmental impact [10]. Precision agriculture [19] [22] has emerged as a crucial solution to address these demands, relying heavily on accurate, large-scale, and up-to-date information regarding land use and crop distribution. Traditionally, acquiring this data required costly and time-consuming manual field surveys, which are impossible to scale effectively at a national or global level.

Over the past decade, Earth Observation (EO) satellites have revolutionized how we monitor agricultural lands. Constellations such as the European Space Agency’s Sentinel-2 provide free, high-resolution multispectral imagery with frequent revisit times (every 5 to 10 days) [7]. By observing how a specific parcel of land reflects light over time [31], it is possible to track the phenological cycle—the biological growth stages—of vegetation. This continuous stream of data allows researchers to identify specific crop types based on their unique growth signatures rather than relying on a single, static snapshot [34].

Simultaneously, the rise of Deep Learning (DL) has provided the computational tools necessary to process this massive influx of satellite data. Early approaches in remote sensing relied heavily on traditional machine learning algorithms and spatial pattern recognition. However, as the field evolved, it became evident that the temporal dimension—how a pixel changes over a year—holds the most predictive power for crop classification. Consequently, sequence-modeling architectures such as Temporal Convolutional Neural Networks (1D-CNNs) [15] [24] and Recurrent Neural Networks (RNNs) [20] have become the standard for agricultural time-series analysis.

Despite these advancements, operationalizing deep learning for crop mapping presents significant difficulties. First, Satellite Image Time Series (SITS) are inherently irregular; optical sensors cannot penetrate clouds, leading to missing data and uneven temporal gaps that disrupt traditional machine learning models. Second, processing high-dimensional spatial-temporal datasets—such as tracking entire 2D image patches over a temporal sequence—creates severe computational and memory bottlenecks, particularly in High-Performance Computing (HPC) environments. Finally, most current state-of-the-art models suffer from “temporal myopia”; they analyze only a single year of weather and phenology, completely ignoring the fundamental agricultural practice of crop rotation, where farmers intentionally alternate crops across multiple years to maintain soil health [4].

This thesis aims to bridge these gaps. To overcome the hardware bottlenecks associated with heavy spatial-temporal grids, this work shifts to a highly optimized 1D pixel-based pipeline. By analyzing the temporal evolution of individual pixels rather than broad spatial patches, the computational load is drastically reduced, enabling rapid prototyping and extensive hyperparameter optimization on an HPC cluster. Building upon this efficient infrastructure, the project establishes robust single-year temporal baselines—specifically 1D-CNNs and Gated Recurrent Units (GRUs)—and compares them against a novel, hybrid multi-year architecture: the Transformer-LSTM. By integrating the self-attention mechanism proven effective on satellite time series [32] [30] with the recurrent memory of an LSTM, the proposed system seeks to answer whether teaching a neural network the historical rules of multi-year crop rotation can significantly outperform models that rely on single-year phenology alone.

1.2 Research Question and Objectives

The goal of this thesis is to explore the efficacy of deep sequence-modeling architectures in classifying agricultural crop types using irregular SITS. Specifically, this research investigates the transition from traditional, single-year phenological analysis to a multi-year, hybrid approach capable of learning

long-term farming practices. This leads to the primary research question of this study:

“How can we efficiently process irregular Satellite Image Time Series for large-scale crop mapping, and does the integration of multi-year crop rotation history via a hybrid Transformer-RNN architecture significantly outperform traditional single-year phenological models?”

To systematically answer this question, the project is structured around four specific objectives, spanning both machine learning infrastructure and architectural design:

- i) **Overcoming I/O and Computational Bottlenecks:** To redesign the data pipeline, shifting from heavy spatial-temporal patches to a highly optimized 1D pixel-based extraction method. This aims to resolve severe High-Performance Computing (HPC) memory constraints and reduce epoch training times, enabling rapid experimentation.
- ii) **Establishing Robust Single-Year Baselines:** To design, train, and tune standard non-hybrid temporal architectures; specifically a 1-Dimensional Convolutional Neural Network (1D-CNN) and a Gated Recurrent Unit Recurrent Neural Network (GRU-RNN). These models serve as the baseline for evaluating standard intra-year phenological classification.
- iii) **Developing a Multi-Year Hybrid Architecture:** To construct a novel Hierarchical Transformer-LSTM model that extends beyond a single harvest cycle. This objective focuses on using self-attention to manage intra-year weather irregularities and an LSTM memory cell to capture inter-year crop rotation dependencies.
- iv) **Automating Hyperparameter Optimization:** To build a robust, fully automated Machine Learning Operations (MLOps) pipeline integrating PyTorch Lightning, Ray Tune, and Optuna. This ensures all models are evaluated fairly at their peak capacity through distributed hyperparameter tuning on an HPC cluster.

1.3 State of the Art

1.3.1 Deep Learning for Precision Agriculture

The paradigm of precision agriculture has fundamentally transformed modern farming management. By transitioning from uniform field-level treatments to highly granular, site-specific interventions, precision agriculture seeks to optimize resource inputs—such as water, fertilizers, and pesticides—while maximizing overall crop yield [19, 22]. This shift relies entirely on the continuous acquisition and analysis of vast amounts of heterogeneous data collected from on site Internet of Things (IoT) sensors, Unmanned Aerial Vehicles (UAVs), and Earth Observation (EO) satellites.

Historically, extracting actionable insights from these diverse data sources depended on traditional machine learning algorithms, such as Random Forests (RF) and Support Vector Machines (SVM) [21, 3]. While effective for specific, localized tasks, these traditional methods are inherently limited by their reliance on manual feature engineering. Researchers were required to construct and select specific vegetation indices or spatial textures beforehand, a process that is highly domain-specific, time-consuming, and difficult to scale across different geographical regions and climatic conditions.

The advent of Deep Learning (DL) has largely overcome these limitations, triggering a major shift in agricultural data analysis [13]. Unlike traditional machine learning, deep neural networks possess the capacity for automatic, hierarchical feature extraction. By ingesting raw or minimally processed data, DL models can autonomously learn complex, non-linear representations directly from the input space. In precision agriculture, this has led to unprecedented breakthroughs across a wide array of applications, including automated weed detection, early disease diagnosis, and precise yield estimation [14].

In the specific context of remote sensing and land cover mapping, deep learning has proven exceptionally proficient at handling the massive dimensionality of Satellite Image Time Series (SITS). By leveraging specialized architectures capable of modeling both spatial hierarchies and temporal dependencies, deep learning models can now accurately map crop distributions at a national scale, outperforming traditional shallow classifiers and forming the foundational technology upon which this thesis is built [16].

1.3.2 Remote Sensing and Sentinel-2 Time Series

The Importance of Spatio-Temporal Dynamics

The European Space Agency’s Sentinel-2 mission has revolutionized agricultural monitoring by providing high spatial resolution imagery with a frequent 5-day revisit time. This creates dense Satellite Image Time Series (SITS) that capture the vital phenological evolution of crops throughout the growing season [29]. However, extracting actionable insights from SITS requires models to navigate a complex time-space tradeoff. Research demonstrates that for crop type classification on Sentinel-2 data, the temporal structure is often richer than the spatial structure; consequently, hybrid models (like recurrent convolutional networks) perform best when the majority of their parameters are allocated to modeling temporal sequences rather than spatial features [29].

Encoding and Sequence Modeling

To effectively process these time series, researchers have explored various representation and sequence-modeling strategies:

- **Image-Based Encodings:** One approach involves transforming 1D time-series data (such as vegetation indices) into 2D image representations—like Gramian Angular Fields or Markov Transition Fields—allowing standard convolutional networks to efficiently extract static and dynamic temporal patterns [6].
- **Recurrent to Transformer Evolution:** While models like Long Short-Term Memory (LSTM) networks have traditionally handled sequential data, modern approaches increasingly rely on Transformer architectures to overcome LSTM limitations with long-range dependencies [36]. In crop mapping, advanced models like the Temporo-Spatial Vision Transformer (TSVIT) utilize purely self-attention mechanisms to process multi-modal SITS, combining optical data (Sentinel-2) with radar (Sentinel-1) to boost classification performance without relying on standard convolutions [9].

Overcoming Noise and Label Scarcity

Despite the rich information in SITS, operational mapping is frequently hindered by cloud cover noise and a severe lack of pixel-level ground truth data. Recent deep learning innovations address these bottlenecks directly:

- **Knowledge-Guided Networks:** To ensure models learn meaningful phenology rather than dataset artifacts, prior domain knowledge can be integrated into the architecture. For instance, the Knowledge-Guided Crop Mapping Network (KGCMNet) forces the model to reconstruct known vegetation indices (like NDVI) during training, which significantly sharpens its spatio-temporal representations [27].
- **Weakly Supervised Learning:** To bypass the need for exhaustive pixel-level annotations, weakly supervised frameworks (e.g., Exact) utilize only image-level labels. These models explore space-time perceptive clues to rectify erroneous semantic biases caused by anomalous temporal clips, pushing weakly supervised performance to nearly 95% of fully supervised baselines [37].
- **Multimodal Pseudo-Labeling:** Another highly scalable solution involves integrating top-down SITS with bottom-up, ground-level field images. By using multimodal data to generate reliable pseudo-labels, models can achieve high accuracy across vast geographical areas while drastically reducing the need for expensive on site data collection [1].

1.3.3 Sequence Modeling: From CNNs to Transformers

The accurate classification of agricultural land relies heavily on the ability to interpret the chronological sequence of crop phenology. As deep learning has evolved, so too have the architectures designed to model these complex temporal dynamics.

Convolutional Neural Networks (CNNs)

Early deep learning approaches for Satellite Image Time Series (SITS) relied heavily on Convolutional Neural Networks (CNNs). While originally designed to extract spatial features from static images [14], CNNs were successfully adapted for temporal sequences via 1-Dimensional (1D) convolutions. By

operating filters across the temporal dimension of a pixel’s profile, 1D-CNNs can efficiently capture localized temporal patterns and phenological shifts, significantly outperforming traditional machine learning classifiers like Random Forests [16]. However, standard CNNs lack an inherent mechanism to model strict, long-term sequential dependencies, as their receptive field is limited by the kernel size.

Recurrent Neural Networks (RNNs), GRUs and LSTMs

To explicitly model the chronological order of agricultural events, Recurrent Neural Networks (RNNs) became the standard. The foundational RNN Encoder-Decoder architecture allowed models to ingest variable-length sequences and maintain a hidden state that summarizes past inputs to inform future predictions [5]. To address the vanishing gradient problem inherent in basic RNNs, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) were widely adopted. These architectures are particularly advantageous in remote sensing, as specialized recurrent layers (such as GRU-D) can be engineered to handle the irregular time gaps, cloud occlusions, and missing values that commonly afflict satellite time series [35].

The Transformer Paradigm

Despite their success, LSTMs process data sequentially, which bottlenecks training parallelization and limits their ability to capture extremely long-range dependencies across a full growing season [36]. The introduction of Transformer architectures marked a paradigm shift in sequence modeling. By dispensing with recurrence and relying entirely on self-attention mechanisms, Transformers evaluate the relationships between all time steps simultaneously, regardless of their distance in the sequence [36]. In modern crop mapping applications, Transformers have demonstrated superior classification accuracy and generalizability compared to 1D-CNNs, especially when handling complex, multi-year, or irregularly sampled temporal features [26].

Hybrid Architectures

Most recently, researchers have sought to combine the localized sequential extraction capabilities of LSTMs with the global receptive field of Transformers. Dual-path and hybrid LSTM-Transformer models process temporal inputs simultaneously, leveraging Bidirectional LSTMs to capture short- and mid-term sequential dependencies, while utilizing Transformer encoders to model long-range spatial and temporal interactions [11]. These hybrid networks mitigate the individual weaknesses of both architectures and represent the current state-of-the-art for decoding the complex spatiotemporal dynamics of Earth observation data [11, 36].

Hybrid Architectures

To maximize predictive performance, researchers have increasingly developed hybrid architectures that integrate multiple sequence-modeling paradigms. Early hybrid networks in remote sensing frequently combined CNNs with RNNs (such as ConvLSTMs or CNN+GRU architectures) to simultaneously extract spatial textures and temporal phenology [29]. However, as the focus has shifted toward capturing highly complex, long-term temporal dynamics, newer approaches have begun merging the localized sequential extraction capabilities of LSTMs with the global receptive field of Transformers. Dual-path and cascaded LSTM-Transformer models process temporal inputs by leveraging Bidirectional LSTMs to capture short- and mid-term sequential dependencies, while utilizing Transformer encoders to model long-range interactions [11]. These advanced hybrid networks mitigate the individual weaknesses of both architectures and represent the current state-of-the-art for decoding the complex spatiotemporal dynamics of Earth observation data [11, 36].

1.3.4 Multi-Year Crop Rotation Modeling

The Importance of Long-Term Agricultural Dynamics

Crop rotation is a fundamental agricultural practice wherein farmers alternate the species of crops grown on a specific plot across consecutive seasons. This strategy is essential for maintaining soil health, preventing nutrient depletion, and breaking pest and disease cycles. Despite its agronomic importance, most remote sensing models suffer from “temporal myopia,” analyzing single-year phenology while ignoring inter-annual historical dependencies.

Traditional and Knowledge-Based Approaches

Early attempts to map multi-year dynamics relied on explicitly injecting prior agronomic knowledge into the classification pipeline. For example, probabilistic models such as Bayesian networks and Conditional Random Fields have been used to encode known crop rotation rules; the outputs of

these networks are then fed into standard classifiers (such as Support Vector Machines) to guide the algorithm and improve early-season mapping accuracy [21, 29]. While effective, these methods require manually constructed rule bases and lack the flexibility to adapt to unexpected crop sequence changes.

Machine Learning and Temporal Transferability

As satellite archives have grown, researchers have increasingly utilized machine learning to track land-use transitions over longer periods. Hierarchical classification workflows (such as tiered Random Forests) have been successfully deployed across multiple years to explicitly track crop rotations, field abandonment, and land-use intensity [3]. However, monitoring annual crop rotations over a decade presents significant challenges regarding temporal transferability; models trained on a specific year often struggle to generalize to past or future years due to shifting weather patterns, varying sensor availability, and inter-annual phenological shifts [26].

The Deep Learning Frontier

While current methods successfully track multi-year changes, they predominantly process each year independently and rely on post-classification comparisons to identify rotations. The next frontier in precision agriculture mapping is the development of deep sequence-modeling architectures that do not require external rulesets, but rather possess the intrinsic memory capacity to learn and predict multi-year agricultural cycles directly from continuous, raw satellite time-series data.

The Deep Learning Frontier

While early deep learning methods successfully tracked multi-year changes, they predominantly processed each year independently and relied on post-classification comparisons to identify rotations. The next frontier in precision agriculture mapping is the development of architectures that possess the intrinsic memory capacity to learn and predict multi-year agricultural cycles directly from continuous, raw satellite time-series data. Recently, researchers have demonstrated significant success in bridging these paradigms by integrating Bayesian modeling—such as Hidden Markov Models (HMMs)—with deep Transformer Encoders to explicitly capture intricate multi-year crop rotation patterns [25]. Building upon these foundations, the focus now shifts toward purely neural hybrid architectures capable of modeling these complex temporal dependencies end-to-end.

1.4 Contributions

This thesis advances the field of agricultural remote sensing by addressing the computational bottlenecks of large-scale mapping and exploring the predictive power of multi-year sequence modeling. The primary contributions of this work are as follows:

- **Adoption of an Optimized 1D Data Pipeline:** To overcome the limitations of memory-intensive 2D spatial patches, a streamlined, 1D pixel-based data representation is utilized. This approach drastically reduces High-Performance Computing (HPC) I/O bottlenecks, enabling the rapid processing and evaluation of massive, multi-year Satellite Image Time Series (SITS).
- **Comprehensive Baseline Benchmarking:** Robust single-year temporal baseline models, specifically a 1-Dimensional Convolutional Neural Network (1D-CNN) and a Gated Recurrent Unit Recurrent Neural Network (GRU-RNN), are developed, trained and thoroughly evaluated. These baselines provide a rigorous standard for intra-year phenological classification against which more complex models can be compared.
- **Novel Multi-Year Hybrid Architecture:** A custom Hierarchical Transformer-LSTM architecture is designed and implemented. This hybrid model uniquely combines the self-attention mechanisms of a Transformer with the recurrent memory of an LSTM to capture both intra-year weather irregularities and inter-year crop rotation dependencies.
- **Integration of Automated MLOps and Distributed Tuning:** A robust Machine Learning Operations (MLOps) workflow, leveraging PyTorch Lightning, Ray Tune, and Optuna, is adapted to automate the training pipeline. This infrastructure facilitates distributed hyperparameter optimization across an HPC cluster, ensuring that all architectures are evaluated fairly and at their maximum potential.

By delivering this end-to-end framework, this thesis not only provides a scalable software solution for operational crop mapping but also offers critical insights into whether neural networks can effectively learn the historical, multi-year rules of agricultural crop rotation.

2 Methodology and System Design

2.1 The INVEKOS Dataset

To evaluate the proposed sequence-modeling architectures, experiments were conducted using the Austrian INVEKOS (Integrated Administration and Control System) dataset. INVEKOS serves as the European Union’s official spatial registry for agricultural parcel identification, providing rich, multi-year ground-truth vector data based on farmers’ declarations [25].

It is important to explicitly state that the data extraction and pre-processing pipeline utilized in this thesis was originally designed and implemented by Perantoni et al. [25]. By adopting their highly optimized dataset structure, this research was able to bypass standard preprocessing bottlenecks and focus entirely on the architectural development of neural network sequence models.

For this project, the spatial boundaries and crop labels were matched with Sentinel-2 Level-1C (L1C) optical imagery, specifically focusing on the Military Grid Reference System (MGRS) tile 33UVP [25]. This provided a continuous, six-year acquisition period spanning from September 2016 to September 2022, encompassing a total of 805 satellite acquisitions with all 13 spectral channels resampled to a 10m resolution [25].

Parameter	Value
Data Source	Sentinel-2 L1C & INVEKOS
Tile	MGRS 33UVP
Timeframe	Sept 2016 – Sept 2022
Total Acquisitions	805
Spatial Resolution	10m
Number of Classes	47
Minimum Parcel Area	400 m ²
Erosion Buffer	15m
Data Splits	60% Train, 20% Val, 20% Test

Table 2.1: Summary of the INVEKOS dataset and extraction parameters.

2.1.1 Data Acquisition (Spatial Patches vs. 1D Pixel Data)

A critical challenge in pixel-based crop mapping is the presence of ”mixed pixels” at the boundaries of agricultural parcels, which contain overlapping spectral signatures from neighboring fields. To mitigate this, a stringent geometric preprocessing pipeline was applied to the INVEKOS shapefiles [25].

Instead of extracting raw 2D spatial patches, spatial erosion buffers were applied to the interior of all field polygons to exclude border pixels. Furthermore, a minimum area threshold of 400 m² was enforced to ensure only sufficiently large and homogeneous parcels were retained [25]. The refined, eroded geometries were subsequently rasterized to align precisely with the Sentinel-2 pixel grid. Finally, to optimize memory overhead, each crop field was collapsed into a single 1D entry representing the spatial mean value of the pixels belonging to that field on every acquisition date [25]. This filtering process yielded an exceptionally clean dataset containing 47 distinct crop type labels [25].

2.1.2 Pre-processing and Handling Irregular Time Series

Handling massive spatial-temporal raster datasets alongside dense vector geometries introduces severe I/O bottlenecks in standard machine learning pipelines. To overcome this, the 1D Sentinel-2 time series data and corresponding ground-truth representations were converted into a streamlined HDF5 (.h5) format, and subsequently into a Lightning Memory-Mapped Database (LMDB) format.

This database architecture enables fast, memory-mapped data reading without requiring the entire multi-gigabyte dataset to be loaded into system RAM. As implemented through the project’s PyTorch Lightning `InvekosDataModule`, the dataset enforces a rigorous stratified random sampling partitioning strategy to ensure unbiased evaluation [25]. The data is split using a standard ratio of 60% for training, 20% for validation, and 20% for testing [25]. This optimized 1D data structure feeds perfectly into data loaders configured with multi-worker pre-fetching, ensuring high throughput across High-Performance Computing (HPC) nodes.

2.2 Single-Year Non-Hybrid Baselines

To establish a rigorous benchmark for multi-year architectures, two single-year sequence modeling baselines were developed and evaluated. These models process intra-year phenological data to provide a comparative standard against which the complexity of hybrid and multi-year models can be justified.

2.2.1 Temporal 1D Convolutional Neural Network (CNN)

The first baseline is a 1-Dimensional Convolutional Neural Network (1D-CNN) designed to extract temporal features by sliding convolutional filters across the time dimension, a methodology that has proven highly effective for satellite time series classification [23]. The **encoder** consists of two sequential convolutional blocks. The initial layer maps the 13 input satellite bands to 64 channels using a kernel size of 3 and a padding of 1 to preserve the temporal sequence length. This is followed by a 1D Batch Normalization layer and an in-place ReLU activation to introduce non-linearity. The second layer further projects the features from 64 to 128 channels, utilizing the same kernel size, padding, normalization, and activation parameters. For **classification**, a point-wise convolution (kernel size of 1) is applied to map the extracted 128-dimensional features to the 47 crop classes. To generate the final field-level classification, the network executes a temporal majority vote by averaging the logit predictions across all time steps in the year.

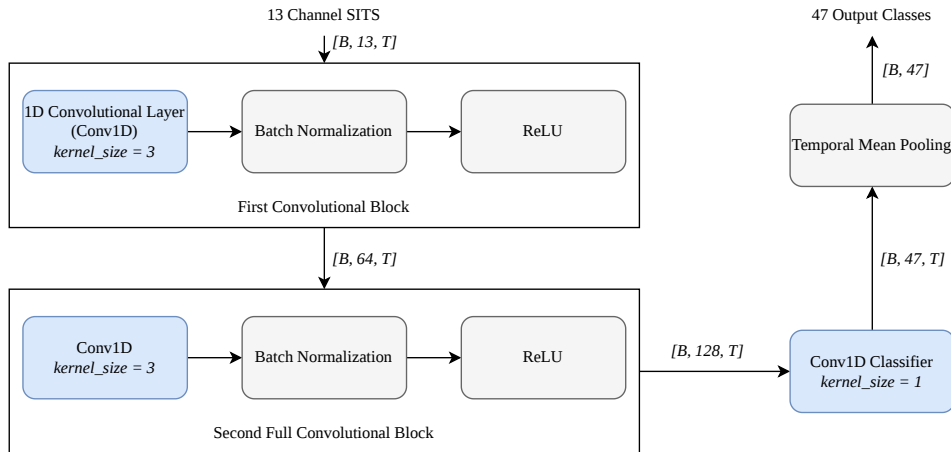


Figure 2.1: Architectural block diagram of the Temporal 1D-CNN baseline.

2.2.2 Temporal Gated Recurrent Unit (GRU RNN)

The second baseline leverages a Gated Recurrent Unit (GRU) [5] to explicitly model the sequential dependencies of agricultural phenology. A **bidirectional GRU** architecture is utilized to process the temporal sequences in both forward and backward directions, yielding better performance than a standard unidirectional RNN. The input tensors are systematically permuted to align the time and

channel dimensions for batch-first sequence processing. Instead of averaging predictions over time, the network isolates the final temporal output step. This final hidden state is fed into a fully connected linear **classifier** to predict the 47 crop classes. Because the network is bidirectional, the input dimension of the final linear classifier is explicitly doubled to accommodate the concatenated forward and backward hidden features. To assess the optimal architectural capacity, the hyperparameter search space tests varying hidden dimensions (128, 256, and 512) and depth configurations (1, 2, and 3 stacked layers).

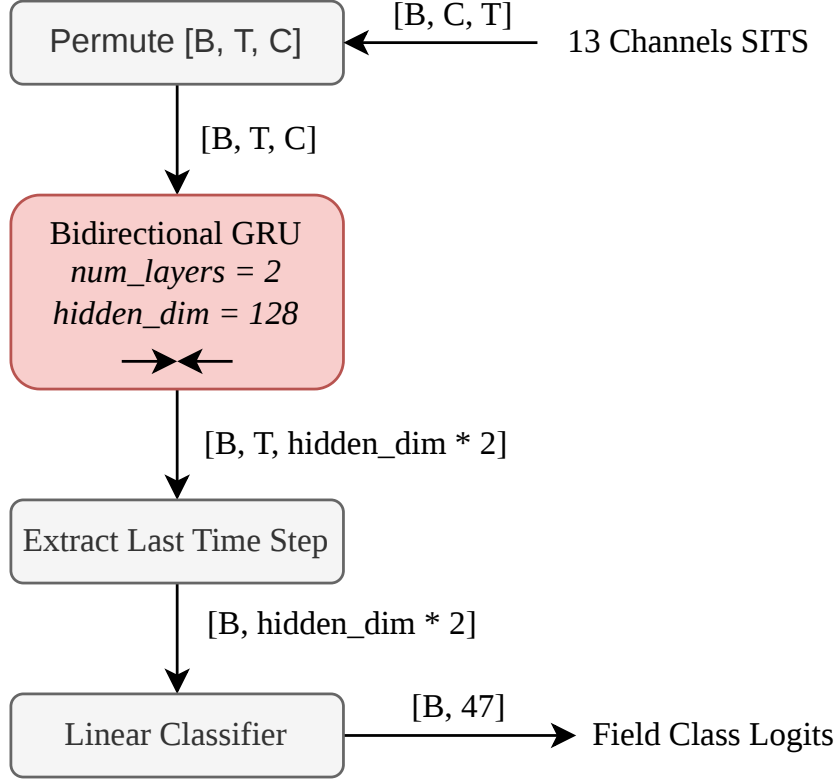


Figure 2.2: Architectural diagram of the Bidirectional GRU baseline.

2.3 Multi-Year Hybrid Architecture

The core contribution of this thesis is the development of a hybrid, multi-scale neural architecture capable of simultaneously modeling intra-year phenological dynamics and inter-year crop rotation patterns. Unlike traditional approaches that treat multi-year classification as independent yearly tasks, this architecture hierarchically processes continuous, multi-year sequences end-to-end.

2.3.1 AgroDays Positional Encoding

The first step in the data pipeline is preparing the raw inputs for sequence modeling. The raw 1D pixel sequence, consisting of 13 spectral channels, is initially projected into a higher-dimensional embedding space (128 dimensions). However, standard sequence models assume that data points arrive at regular, evenly spaced intervals. Agricultural satellite time series are inherently irregularly sampled due to cloud cover and varying satellite revisit times. Using standard sequential positional encodings would falsely inform the model that all observations are equidistant. To address this, the architecture utilizes a custom **AgroDays Positional Encoding** module. Instead of simply indexing observations as sequential integer steps, this module adapts the standard sinusoidal positional encoding [32] by

scaling the frequencies using the exact agricultural day of the acquisition. By calculating the encoding angles based on the absolute temporal position of the specific acquisition day, the model inherently understands the varying, irregular time gaps between observations, ensuring the network accurately aligns the spectral data with the correct stage of the crop’s phenological cycle.

Algorithm 1 AgroDays Positional Encoding

Require: Flattened input sequence $X \in \mathbb{R}^{(B \cdot Y) \times T \times d_model}$

Require: Agricultural days matrix $pos \in \mathbb{R}^{(B \cdot Y) \times T}$

Ensure: Position-encoded sequence $X_{enc} \in \mathbb{R}^{(B \cdot Y) \times T \times d_model}$

- 1: $div_term \leftarrow \exp \left(\text{arange}(0, d_model, 2) \cdot \frac{-\ln(10000.0)}{d_model} \right)$
 - 2: $pos_expanded \leftarrow \text{unsqueeze}(pos, \text{dim} = -1)$
 - 3: $angles \leftarrow pos_expanded \times div_term$
 - 4: $PE \leftarrow \text{zeros.like}(X)$
 - 5: $PE_{::,0::2} \leftarrow \sin(angles)$ ▷ Apply sine to even indices
 - 6: $PE_{::,1::2} \leftarrow \cos(angles)$ ▷ Apply cosine to odd indices
 - 7: $X_{enc} \leftarrow X + PE$ ▷ Inject temporal signal into embeddings
 - 8: **return** X_{enc}
-

2.3.2 The Hierarchical Transformer-LSTM Model

Once the position-aware embeddings are generated, the network processes the temporal scales using a two-stage hierarchical model. In the first stage, the objective is to understand the intra-year phenological growth cycle. To optimize computational efficiency, the multi-year input tensor is reshaped, flattening the batch and year dimensions together. This allows an Intra-Year Transformer Encoder to process every year independently and in parallel across the GPUs. Driven by Multi-Head Self-Attention [32], the Transformer evaluates the entire yearly sequence simultaneously, effectively capturing complex growth stages without being hindered by vanishing gradients [28].

Algorithm 2 Hierarchical Transformer-LSTM Forward Pass

Require: Multi-year input tensor $X \in \mathbb{R}^{B \times Y \times C \times T}$

Require: Agricultural days tensor $pos \in \mathbb{R}^{B \times Y \times T}$

Ensure: Crop class logits $Y_{pred} \in \mathbb{R}^{B \times n_classes \times Y}$

- 1: **Stage 1: Intra-Year Processing (Transformer)**
 - 2: $X \leftarrow \text{permute}(X, [0, 1, 3, 2])$ ▷ Target shape: $[B, Y, T, C]$
 - 3: $X_{flat} \leftarrow \text{reshape}(X, [B \cdot Y, T, C])$ ▷ Flatten for independent yearly processing
 - 4: $pos_{flat} \leftarrow \text{reshape}(pos, [B \cdot Y, T])$
 - 5: $X_{emb} \leftarrow \text{LinearProjection}(X_{flat})$
 - 6: $X_{emb} \leftarrow \text{AgroDaysPositionalEncoding}(X_{emb}, pos_{flat})$
 - 7: $T_{out} \leftarrow \text{TransformerEncoder}(X_{emb})$ ▷ Self-Attention over time steps
 - 8: $Y_{emb} \leftarrow \text{mean}(T_{out}, \text{dim} = 1)$ ▷ Temporal pooling: $[B \cdot Y, d_model]$
 - 9: **Stage 2: Inter-Year Processing (LSTM)**
 - 10: $Y_{seq} \leftarrow \text{reshape}(Y_{emb}, [B, Y, d_model])$ ▷ Restore sequence of years
 - 11: $L_{out}, (h_n, c_n) \leftarrow \text{BidirectionalLSTM}(Y_{seq})$ ▷ Model crop rotations
 - 12: **Stage 3: Classification**
 - 13: $logits \leftarrow \text{LinearClassifier}(L_{out})$ ▷ Map to $n_classes$
 - 14: $Y_{pred} \leftarrow \text{permute}(logits, [0, 2, 1])$ ▷ Final shape: $[B, n_classes, Y]$
 - 15: **return** Y_{pred}
-

The time steps for each year are then mean-pooled, collapsing the complex intra-year phenology into a single, dense feature vector per year. In the second stage, these extracted yearly embeddings are restored into their chronological, multi-year sequence. This sequence is then fed into an Inter-Year Long Short-Term Memory (LSTM) network [12] to transition from intra-year phenology to inter-year

crop dependencies.

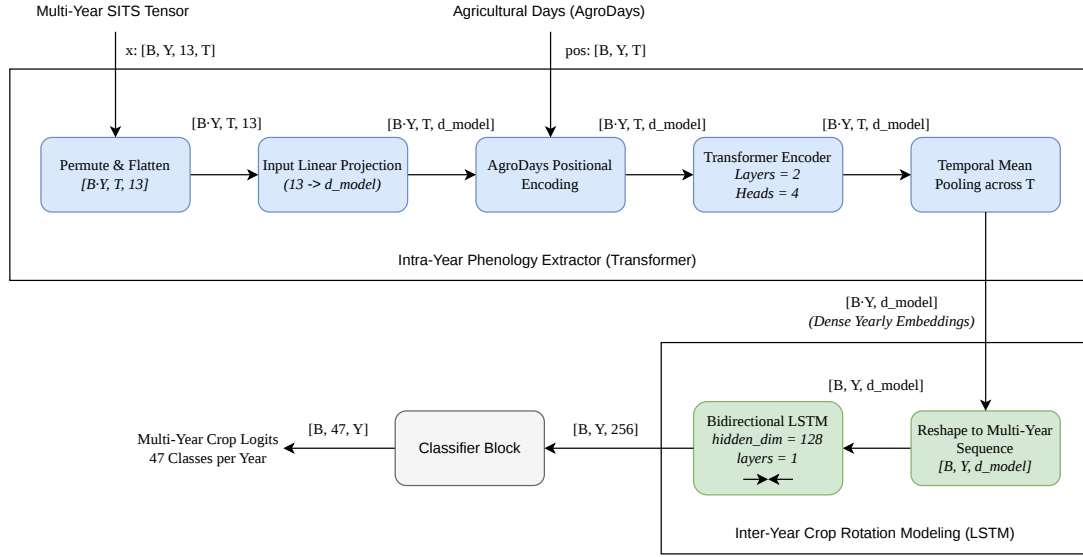


Figure 2.3: Hierarchical overview of the Transformer-LSTM architecture, demonstrating the fusion of intra-year and inter-year components.

2.3.3 Capturing Long-Term Crop Rotations

The primary motivation for integrating the LSTM in the second stage is its exceptional ability to model strict, long-term sequential rules. Traditional crop mapping architectures primarily focus on extracting phenological features within a single growing season, discarding a vast amount of predictive temporal context. However, agricultural practices are heavily governed by multi-year crop rotation cycles designed to maintain soil health and comply with policies. To explicitly capture these long-term crop rotations, a bidirectional LSTM configuration is employed. This enables the network to analyze the rotation history in both the forward and backward temporal directions, effectively predicting a crop based on what was planted in previous years as well as what will be planted in subsequent years. By analyzing this bi-temporal history, the network learns deep inter-year transition probabilities. Finally, the bidirectional output—representing this deeply fused, multi-year context—is passed to a fully-connected classification block. Because the LSTM is bidirectional, the sequence dimension is explicitly doubled to accommodate the concatenated forward and backward hidden states. These deep features are passed through a ReLU activation and a dropout regularization layer (set to 10%) to prevent overfitting. A final linear projection maps the processed features into the unnormalized logit scores for the 47 target crop classes, outputting a highly informed prediction for each year in the sequence.

2.4 Infrastructure and Optimization

Training deep temporal models on massive satellite datasets requires a robust, scalable infrastructure. To ensure reproducibility, efficient hardware utilization, and systematic hyperparameter selection, the experimental framework was constructed using PyTorch Lightning [8], Ray Tune [17], and MLflow [33].

2.4.1 PyTorch Lightning Training Pipeline

The training logic for all models was encapsulated within the PyTorch Lightning framework [8], abstracting code engineering while ensuring multi-GPU scalability. Several optimizations were implemented at the pipeline level to stabilize training and maximize throughput:

Mixed Precision Training: To optimize GPU memory and accelerate tensor operations, the trainer was configured to use 16-bit mixed precision (`16-mixed`). This allows the models to perform computationally heavy operations in half-precision while maintaining full precision for gradient

accumulations, maximizing the throughput of the underlying hardware.

Loss Formulation: The networks were trained using a standard Cross-Entropy Loss function. Crucially, an `ignore_index` parameter set to 255 was passed to the loss function. This ensured that any padded sequences or years dropped during data augmentation did not contribute to the loss gradients or skew the accuracy metrics.

Optimization and Regularization: Optimization was governed by the AdamW algorithm [18]. Unlike the standard Adam optimizer, AdamW correctly decouples weight decay from the adaptive learning rate updates, providing superior regularization [18]. To prevent unnecessary computational expenditure on converging models, an Early Stopping callback was implemented, terminating training if the validation accuracy (`val_acc`) failed to improve for 10 consecutive epochs.

2.4.2 Distributed Hyperparameter Tuning with Ray Tune and Optuna

Neural network sequence models are highly sensitive to their hyperparameters. To systematically search for the optimal configuration, a distributed tuning loop was orchestrated using Ray Tune [17], integrated with the Tree-structured Parzen Estimator (TPE) sampling algorithm from Optuna [2].

The tuning process was executed with a budget of 30 total trials. The Optuna sampler was initialized with 20 completely random startup trials to thoroughly explore the boundaries of the parameter space before transitioning to a Bayesian optimization strategy. To maximize hardware efficiency, Ray Tune was configured to run 3 trials concurrently. During this exploratory phase, each trial was trained for a truncated duration of 5 epochs.

Algorithm 3 Distributed Ray Tune and Lightning Orchestration

Require: Search space $\mathcal{S}$, Trials $N = 30$, Concurrent Workers $W = 3$

Ensure: Best configuration $config_{best}$ and evaluated Test Metrics

```

1: function METAPARAMETERSTUNING( $\mathcal{S}$ )
2:    $\mathcal{D} \leftarrow$  Initialize Optuna TPE Sampler(startup_trials = 20)
3:    $results \leftarrow$  RayTuneRun(
4:     train_fn = LightningTrainer,
5:     search_space =  $\mathcal{S}$ ,
6:     num_samples =  $N$ ,
7:     max_concurrent =  $W$ ,
8:     scheduler =  $\mathcal{D}$ 
9:   )
10:   $config_{best} \leftarrow$  GetBestConfig( $results$ , metric = 'val_acc', mode = 'max')
11:  return  $config_{best}$ 
12: end function

13: Execution:
14:  $config_{best} \leftarrow$  METAPARAMETERSTUNING( $\mathcal{S}$ )  $\triangleright$  Run 5 epochs per trial
15:  $Model_{final} \leftarrow$  InitializeModel( $config_{best}$ )
16:  $Trainer_{final} \leftarrow$  LightningTrainer(max_epochs = 40, early_stopping = 10)
17:  $Trainer_{final}.fit(Model_{final})$ 
18:  $TestMetrics \leftarrow$   $Trainer_{final}.test(Model_{final})$ 

```

The continuous search space included a log-uniform distribution for the learning rate (bounded between $1e-5$ and $1e-2$) and the weight decay (bounded between $1e-7$ and $1e-3$). Discrete grid-search choices were provided for the batch size (8, 16, or 32). Furthermore, architecture-specific parameters were dynamically tuned:

- For the **GRU RNN**, the search space tested hidden dimensions of 128, 256, and 512, alongside variable network depths of 1, 2, or 3 stacked layers.
- For the **Transformer-LSTM**, the core embedding dimension (`d_model`) was tested across 128, 256, and 512.

Every trial automatically logged its hyperparameters and performance metrics (including training loss, validation loss, overall accuracy, and F1-score) to a local MLflow database [33]. After the tuning phase concluded, the configuration that yielded the maximum validation accuracy was extracted. This optimal configuration was then used to instantiate a final model, which was trained from scratch for 40 epochs and rigorously evaluated on the held-out test set.

Hyperparameter	Distribution/Values	Applies To
Learning Rate	Log-Uniform [1e−5, 1e−2]	All Models
Weight Decay	Log-Uniform [1e−7, 1e−3]	All Models
Batch Size	{8, 16, 32}	All Models
GRU Hidden Dim	{128, 256, 512}	GRU RNN
GRU Layers	{1, 2, 3}	GRU RNN
Transformer <i>d_model</i>	{128, 256, 512}	Transformer-LSTM

Table 2.2: Hyperparameter search space defined for Ray Tune optimization.

3 Experiments and Discussion

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3.1 Experimental Setup

3.1.1 HPC Environment and I/O Bottleneck Mitigation

3.1.2 Evaluation Metrics

3.2 Baseline Results (Non-Hybrid Models)

3.2.1 Temporal CNN Performance

3.2.2 Temporal GRU RNN Performance

3.3 Proposed Architecture Results (Hybrid Model)

3.3.1 Hierarchical Transformer-LSTM Performance

3.4 Discussion

3.4.1 Single-Year vs. Multi-Year Capabilities

3.4.2 The Impact of Class Imbalance

4 Conclusions and Future Work

4.1 Conclusions

4.2 Future Work

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